

InsightChain: Optimized Chain-of-Insight Analytics for LLM-driven Data Visualization

Hanya Sun* Chen Zhang*[†] Sheng Liang Yongyue Zhang Yong Liu
Huawei Technologies Co., Ltd.
zhang.chen4@huawei.com

Abstract

Large language models (LLMs) are increasingly used for automated data visualization, yet existing approaches often frame visualization generation as a single-step mapping from user query to figure or code, overlooking the iterative analytical reasoning process of expert analysts. We present InsightChain, a four-stage visualization prompting pipeline (Explore–Focus–Test–Present) that emulates expert analytical workflows, together with VG-COPRO, a vision-guided automatic prompt optimization (APO) method adapted to jointly optimize such multi-stage, executable pipelines. To address the evaluation gap for complex data visualization, we introduce the Insight Progression Metric (IPM), a rubric combining four text-based dimensions with a vision-based dimension. We assess IPM through a 100-chain human pilot and an expanded 300-chain agent-based evaluation spanning all ten domains. Experiments on public datasets show that InsightChain consistently outperforms competing prompting baselines. Existing APO methods fail to yield consistent gains on this multi-stage task, whereas VG-COPRO improves performance in both in-domain and cross-domain settings.

1 Introduction

Large language models (LLMs) have significantly advanced automated data visualization (Dibia, 2023; Tian et al., 2025). However, most existing approaches treat visualization as a single-step mapping from query to executable code or rendered figure, bypassing intermediate validation (Cheng et al., 2023). This one-shot approach contradicts expert sensemaking workflows (Pirulli and Card, 2005), which are inherently iterative: analysts explore broad data distributions (Tukey, 1977), formulate specific hypotheses (Heer and Shneiderman, 2012), and refine visual encodings (Munzner, 2014)

to synthesize reliable insights (Sacha et al., 2014). By compressing this multi-stage reasoning into a single run, current LLM strategies often yield superficial visualizations that lack statistical rigor and fail to uncover deep insights in complex real-world data.

To bridge this gap, we introduce InsightChain, a multi-stage prompting pipeline that emulates expert workflows through four stages: Explore, Focus, Test, and Present. Because these stages are interdependent, optimizing their prompts requires feedback that accounts for downstream execution and the rendered output. We therefore study how existing automatic prompt optimization (APO) methods transfer to this setting and introduce VG-COPRO, a task-adapted method that combines COPRO’s per-predictor search with OPRO-style proposal and vision-aware, pipeline-level evaluation.

Evaluating this form of multi-stage analysis is difficult because a table may support multiple valid analytical directions, leaving no unique reference insight or visualization, while final-output metrics cannot determine whether intermediate hypotheses are grounded in executed evidence. We therefore introduce the Insight Progression Metric (IPM), a rubric tailored to multi-step visualization tasks, for evaluating both the analytical trajectory and the final visualization. Our experiments address four questions: (1) how well IPM agrees with human evaluation, and how robustly it behaves across repeated and cross-family judgments; (2) whether the chain pipeline improves over non-chain prompting baselines across task models; (3) whether VG-COPRO yields consistent gains over the unoptimized chain and existing APO methods; and (4) whether the optimized prompts transfer when the Present stage is rendered by a text-to-image model. Our contributions are three-fold:

- We propose **InsightChain**, an execution-grounded prompting pipeline for automated

*Co-first Authors

[†]Corresponding Author

data visualization that follows four stages of expert sensemaking: Explore, Focus, Test, and Present.

- We systematically study how existing APO methods transfer to multi-stage, executable data analysis and introduce **VG-COPRO**, a task-adapted optimizer that combines dimension-level diagnostic feedback, a leave-one-domain-out robustness objective, and vision-aware evaluation.
- We introduce the **Insight Progression Metric (IPM)**, a rubric for evaluating multi-step visualization tasks, assess its agreement with human ratings, and examine its repeatability and cross-family robustness through agent-based judgments. Experiments across ten domains demonstrate the effectiveness of InsightChain and VG-COPRO, including transfer to a text-to-image renderer.

2 Related Work

2.1 LLM-driven visualization

Early systems for natural-language-driven chart generation typically treat the task as a single-pass mapping from query to code or chart specification. For instance, while LIDA (Dibia, 2023) introduces an LLM-backed goal-then-code pipeline, its goal generation and code generation remain decoupled, exchanging no execution-grounded context. Similarly, ChartGPT (Tian et al., 2025) relies on a single prompt conditioned on user queries and schemas, and Prompt4Vis (Li et al., 2025a) improves few-shot selection but rigidly retains the one-shot input-output structure. Other systems, such as Chat2VIS (Maddigan and Susnjak, 2023) and multi-agent extensions like Data-to-Dashboard (Zhang and Elhamod, 2025), add role decomposition, yet still lack an execution-grounded insight-progression objective. These systems primarily address data-plus-intent-to-visualization and remain the closest executable visualization baselines available. InsightChain instead begins without a predefined analytical intent and links iterative reasoning with execution feedback; our comparison therefore measures their practical performance under this open-ended setting rather than assuming identical task formulations.

2.2 Insight discovery and analytical report generation

InsightBench (Sahu et al., 2025) and DiscoveryBench (Majumder et al., 2025) evaluate generated textual insights or hypotheses against reference discoveries. DAComp (Lei et al., 2026) evaluates analytical reports conditioned on predefined business questions, while Multimodal DeepResearcher (Yang et al., 2026) begins with a topic and web retrieval to produce an interleaved multimodal report. These tasks are related to InsightChain in analytical breadth but differ in their inputs and outputs: InsightChain begins with a single table and no predefined intent, and requires executable exploration culminating in a visualization. We therefore discuss these systems as related tasks but do not directly compare scores obtained under their incompatible evaluation protocols.

2.3 Visualization quality metrics

Existing evaluation metrics for LLM-driven data visualizations generally fall into three categories, each exhibiting structural limitations that motivate the design of our IPM rubric.

Specification-based metrics, such as nvBench (Luo et al., 2021) and Prompt4Vis (Li et al., 2025a), judge a chart as correct only if its Vega-Lite specification matches a reference exactly, penalizing equally valid visualizations with different encodings. Text-based metrics, like Chart-to-Text (Kantharaj et al., 2022), compare generated captions with references using BLEU; they tolerate surface differences but ignore whether the chart itself is accurate or renderable. Vision-language metrics, including CLIP similarity (Radford et al., 2021), LIDA (Dibia, 2023), and chart-VQA adaptations, evaluate rendered images directly, yet mainly capture image-text alignment rather than data fidelity, so charts with misleading axes or hidden aggregations may score higher than honest representations.

Our IPM addresses these limitations by evaluating both the intermediate analytical reasoning and the visual fidelity of the outputs. A detailed property comparison is provided in Table 8 (Appendix A.3).

2.4 Automatic prompt optimization

Recent work on APO spans several algorithmic paradigms: Bayesian search (MIPROv2, Opsahl-Ong et al., 2024); coordinate ascent (COPRO, Khat-tab et al., 2024); LLM-as-optimizer (OPRO, Yang

et al., 2024); evolutionary search (EvoPrompt, Guo et al., 2024); and fixed structured-template scaffolds (Plan-and-Solve, Wang et al., 2023). These paradigms are typically evaluated on text-only classification or QA benchmarks (HotpotQA, GSM8K, IRIS) with large validation sets and discrete correctness labels. However, applying them to multi-step visualization agents reveals two critical structural limitations: they optimize prompts in isolation rather than as a pipeline, and they rely solely on text-based signals, remaining blind to the visual fidelity of the rendered figure. These limitations motivate our design of VG-COPRO, a vision-guided, pipeline-aware approach.

3 Methodology

3.1 The InsightChain Pipeline

Task formulation. Given a single tabular dataset D without a predefined analytical question or visualization intent, the system must identify a worthwhile analytical direction, test the resulting hypothesis through executable analysis, and communicate the supported finding in a visualization. The output is therefore a multi-stage reasoning-and-execution trace together with a final figure, rather than a chart generated from a user-specified intent. InsightChain implements this task through a four-stage pipeline and uses VG-COPRO to optimize its stage-specific prompts.

Visualization analysis is formalized as a structured four-stage pipeline for iterative reasoning, $S = (t_1, t_2, t_3, t_4)$ ¹. At each stage t , a predictor π_t , implemented as a DSPy chain-of-thought module with a typed signature, produces a structured output $o_t = \pi_t(x_t; \theta_t)$, where x_t denotes the input context and θ_t denotes the stage-specific prompt.

Each output o_t contains an executable Python program py_t , which is executed by a sandboxed executor $\mathcal{E}$ with retry budget $B = 2$:

$$e_t = \mathcal{E}^{(B)}(py_t, D) \quad (1)$$

where D denotes the input table. Upon execution, the environment returns an execution state $e_t = (\text{stdout}_t, \text{figs}_t, \text{err}_t)$. Here, stdout_t captures

¹The four stages t_1, t_2, t_3 , and t_4 correspond to the Explore, Focus, Test, and Present phases, respectively. Explore corresponds to Tukey’s exploratory data analysis (Tukey, 1977); Focus to the hypothesis-formulation step of interactive analysis (Heer and Shneiderman, 2012); Test to standard statistical validation; and Present to the encoding-and-communication phase of visualization design (Munzner, 2014).

the standard textual output, figs_t denotes the rendered visualizations, and err_t logs runtime errors. If $\text{err}_t \neq \emptyset$, the error trace is fed back to the predictor π_t , guiding the regeneration of o_t and thus forming a self-correcting execution loop.

Downstream stages condition on prior outputs and execution traces. We collect the record of each completed stage into a running context $C_{<t} = (o_1, e_1, \dots, o_{t-1}, e_{t-1})$, with $C_{<1} = \emptyset$. The input to every predictor is uniformly

$$x_t = (\sigma(D), C_{<t}), \quad (2)$$

where $\sigma(D)$ is a fixed dataset profile (column names, dtypes, summary statistics, and a five-row sample). The pipeline’s final deliverable is $(o_4.\text{FINAL_INSIGHT}, \text{figs}_4)$; the stage records $\{(o_t, e_t)\}$ serve as the evaluation basis for IPM and the optimization signal for VG-COPRO. The prompt θ_t operates on top of the typed signature. We present the Explore instruction verbatim below to illustrate the prompt format (full-stage prompts are provided in Appendix A.5):

“Given a dataset summary and sample, generate Python visualization code that broadly explores the data landscape with 2–3 overview charts. Identify distributions, relationships, temporal patterns, and outliers. Print key observations and identify non-obvious patterns worth investigating.”

3.2 Insight Progression Metric (IPM)

Before diving into the details of prompt optimization, we first define the optimization objective used to quantify improvements in insight visualization. We propose the Insight Progression Metric (IPM), a weighted linear combination of five quality dimensions computed from the chain’s source code, intermediate insights, and rendered figures:

$$\text{IPM}(c) = \sum_{d \in \mathcal{D}} w_d \cdot \phi_d(c), \quad (3)$$

where $\mathcal{D} = \{\text{EXPLORATION BREADTH, FOCUS RELEVANCE, HYPOTHESIS RIGOR, INSIGHT COMMUNICATION, VISUAL QUALITY}\}$, $\phi_d \in [1, 10]$ is the LLM-as-judge score, and $w = (0.20, 0.25, 0.20, 0.25, 0.10)$ ².

The first four dimensions correspond to the four stages of InsightChain and are evaluated by a text-based judge using the generated code and insight

²The weights were specified manually from the task requirements. Section 5.1 evaluates uniform, perturbed, fully random, and data-fitted alternatives and shows that the principal conclusions are stable across these choices.

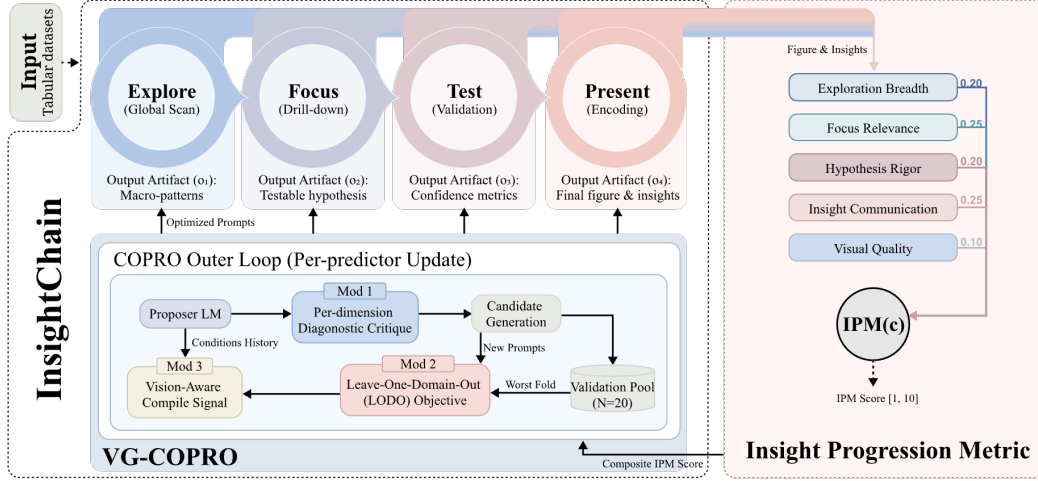


Figure 1: **System overview.** **InsightChain** (left) consists of (i) a four-stage Explore→Focus→Test→Present pipeline, ingesting tabular datasets and emitting an output artefact (α_t) at each stage that conditions the next; and (ii) VG-COPRO, the prompt optimizer that wraps an OPRO-style proposer LM in COPRO’s per-predictor outer loop, with three modifications selecting candidate prompts on a 20-example validation pool by composite IPM score. **IPM** (right): a five-dimensional rubric with weights (0.20, 0.25, 0.20, 0.25, 0.10) scored by an LLM-as-judge, producing a composite score that serves both as the evaluation metric and as the VG-COPRO compile signal.

descriptions from each stage. **VISUAL QUALITY** is assessed by a vision-language model using the figure rendered in the Present stage.

VISUAL QUALITY contributes 10% to the composite IPM score; it complements the four text-based dimensions rather than determining the overall evaluation on its own. Within this dimension, the central criterion is insight embodiment: whether the visualization effectively communicates the final finding through an appropriate chart type and elements such as annotations, reference lines, and statistical overlays, rather than merely displaying the data correctly. A faithful but uninformative chart therefore receives only a moderate or low score. The score is capped at 3 if either (1) the figure misrepresents the data or (2) fundamental legibility failures trigger the display gate. Detailed gate conditions appear in the annotator handbook (Appendix A.8). If the final figure fails to render, the score is set to $\phi_5 = 1$. All results in § 5 use this IPM formulation, which also provides the optimization signal for the APO methods.

3.3 Prompt Optimization Paradigms

We study automatic prompt optimization (APO) for InsightChain, treating proprietary LLMs as fixed-weight black-box APIs. First, we examine how existing APO methods interact with our four-stage InsightChain framework. We recompile the same InsightChain using five representative methods

spanning the current APO taxonomy: MIPROv2 (Opsahl-Ong et al., 2024), COPRO (Khattab et al., 2024), OPRO (Yang et al., 2024), EvoPrompt (Guo et al., 2024), and Plan-and-Solve (Wang et al., 2023). These algorithms differ on two axes, search target and proposer mechanism, as summarised in Table 9 of Appendix A.4.

Because OPRO, EvoPrompt, MIPROv2, and Plan-and-Solve were originally designed for single-prompt optimization, while InsightChain consists of four predictors, we adapt all five methods using COPRO’s per-predictor outer-loop strategy. Specifically, the instruction of one stage is optimized at a time while the remaining stages are kept fixed. Only the inner proposal-generation mechanism varies for different APOs. Moreover, Plan-and-Solve serves as a no-search baseline. Its proposer slot is replaced with a fixed plan-then-solve template applied once per predictor, allowing us to isolate the contribution of iterative search beyond a carefully designed template.

3.4 VG-COPRO: a vision-guided APO

When adapted to InsightChain using the common per-predictor outer loop, existing APO methods yield no consistent improvement across both evaluation splits. This setting presents three task-specific difficulties: a prompt change at one stage propagates through downstream execution, average validation performance may not transfer across do-

mains, and text-only feedback does not assess the rendered figure. We therefore develop VG-COPRO as a task-adapted optimizer built on COPRO and OPRO, rather than as a new general-purpose APO framework. Each candidate stage prompt is assessed by executing the complete pipeline, allowing optimization to account for downstream effects, cross-domain robustness, and visual feedback.

VG-COPRO is built upon the COPRO outer loop for iterative per-predictor optimization. For the inner loop, we adopt OPRO’s proposer, since it is the only method achieving a positive in-domain gain and supporting non-scalar feedback in Table 1. Based on this structure, we introduce three independent modifications to VG-COPRO.

Mod 1: per-dimension diagnostic critique. Instead of a single scalar reward, the proposer LM receives the full five-dimensional IPM breakdown together with the weakest-performing dimension. This converts the optimization feedback into a structured diagnostic signal, extending the textual-feedback paradigm of Yuksekgonul et al. (2025). **Mod 2: leave-one-domain-out (LODO) objective.** Fitness is defined as the minimum IPM across the five disjoint in-domain folds, rather than their mean. This formulation acts as a worst-case robustness objective, inspired by Group DRO (Sagawa* et al., 2020), and better aligns the optimization signal with cross-domain generalization. **Mod 3: vision-aware compile signal.** Every candidate prompt is evaluated using the full 5-dimensional IPM, including the vision-judge assessment. The full pseudocode is shown in Algorithm 1 (Appendix A.5).

4 Experimental Setup

Datasets. We aggregate 285 tabular datasets spanning 10 domains from seven public sources. Under a leave-five-domains-out protocol, five domains supply 150 datasets for compilation (train $n=95$, validation $n=20$, and in-domain test $n=35$), while the other five domains provide the cross-domain test set ($n=135$). Each n denotes an independent analysis task associated with a distinct dataset, rather than a table row. The median dataset contains approximately 6,000 rows; the in-domain and cross-domain test sets contain approximately 550K and 2.09M records, respectively. Overall, 91.9% of the datasets come from real public sources, with a small synthetic-template fallback for underrepresented demographics and en-

ergy domains. Source composition, domain counts, and split sizes are detailed in Appendix A.10.

IPM judges. The primary text and visual judge used for IPM scoring is Gemini 3 Flash, with inference temperature set to 0. To assess potential same-family judge bias, GPT-4.1 and Qwen-VL-Max re-score a stratified subsample of $n = 25$ chains per condition across three baselines and five APO methods using the same rubric.

Task models and other metrics. Chain generation and baselines run on Gemini 3 Flash (primary; temperature 0.3, 10,000-token output limit) and on GPT-4.1 for cross-model replication. Generated code executes in a sandboxed subprocess (60 s timeout, up to two retries). For the auxiliary visualization metrics, we use OpenCLIP ViT-L/14 for CLIP similarity, SacreBLEU for Chart-to-Text BLEU, and the installed lida v0.0.14 package for the LIDA VLM evaluator.

APO compilation. All APOs share a single train/val pool ($n_{\text{val}} = 20$) drawn from the five training domains, with a per-predictor budget of three compile iterations. VG-COPRO evaluates each candidate using five leave-one-domain-out folds constructed exclusively from the training domains: finance, healthcare, social science, climate, and sports. The optimizer never accesses the five cross-domain test domains: education, transportation, e-commerce, energy, and demographics.

Text-to-Image transferability. We also evaluate the transferability of InsightChain prompts optimized with VG-COPRO to Text-to-Image (T2I) models. In this setting, the Present stage is rendered using gemini-3.1-flash-image-preview, while the three preceding stages, datasets, IPM weights, and evaluation judges remain unchanged.

5 Results

5.1 IPM Validation

We assess IPM using complementary tests of controlled-band discriminability, human agreement, weight sensitivity, judge self-consistency, dimension overlap, and cross-family robustness. Extended analysis on agent-based repeatability, expert-audited criteria, and close-pair discrimination are stated in Appendix A.2.

For the controlled-band analysis, we construct a corpus of 100 chains from 20 held-out datasets, where each dataset is evaluated under five prompt

regimes corresponding to different quality levels: HIGH, MEDIUM-A, MEDIUM-B, LOW, and RANDOM-PERMUTATION. The HIGH regime encourages expert-level statistical analysis and publication-ready visualizations, while MEDIUM-A and MEDIUM-B use the same analyst-level prompt under different random seeds to measure robustness against stochastic variation. The LOW regime contains minimal plotting instructions without insight requirements, and RANDOM-PERMUTATION serves as a content-broken control with unlabeled and semantically meaningless figures. This paired 20×5 design ensures that all regime comparisons are controlled at the dataset level. The full system prompts and the chain-generation script are documented in Appendix A.9.

Discriminability. The five-dimensional IPM clearly distinguishes the controlled quality bands (Figure 2). A one-way ANOVA conducted on the aggregate IPM scores across the four real-quality groups (HIGH, MEDIUM-A, MEDIUM-B, and LOW) yields $F(3, 96) = 153.1$ with $p < 10^{-36}$, indicating strong statistical separation between quality levels. As intended, IPM does not significantly distinguish MEDIUM-A from MEDIUM-B, which use the same prompt and differ only in random seed; it separates the remaining adjacent quality levels with Cohen’s $d > 1.8$. Each individual IPM dimension also independently separates the quality bands with statistical significance ($p < 10^{-4}$).

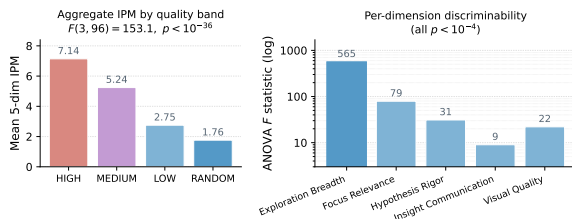


Figure 2: IPM discriminability on the 100-chain pilot. **Left:** aggregate 5-dim IPM by quality band. **Right:** per-dimension ANOVA F statistic (all $p < 10^{-4}$).

Human agreement. Two members of the author team, trained using the annotator handbook and blind to both the chain source condition and the IPM scores, evaluated the 100-chain pilot (Appendix A.8). Among the $n = 97$ pilot chains with complete overall-quality ratings, the IPM composite score shows strong agreement with the human ratings, achieving Spearman $\rho = 0.88$ (Table 5 in Appendix A.1). IPM assigns systematically lower

scores than the human annotators on the four text-based dimensions while closely matching them on VISUAL QUALITY. This difference primarily reflects a shift in scoring scale rather than disagreement in ordering, as IPM and the human annotators rank the quality regimes consistently.

Weight sensitivity. Because the IPM weights are manually specified, we test whether the conclusions depend on this choice. Uniform weights leave the ranking of all nine methods unchanged (Kendall’s $\tau = 1.0$), with all significance conclusions preserved. Across 500 small perturbations around the original weights, the rankings remain stable, with mean $\tau = 0.98$ in-domain and 0.95 cross-domain; VG-COPRO remains the top-performing method in almost every case. Under 500 completely random weight configurations, the advantage of chain-based methods over single-shot methods remains consistently significant. Data-fitted weights achieve nearly identical agreement to the manual weights ($\rho = 0.894$ versus 0.882). These results indicate that the central conclusions are not determined by the exact manual weighting.

Self-consistency. Because the main experiments use temperature-0 decoding with prompt caching, repeated evaluations are effectively deterministic. To estimate residual variability in the judges, we therefore conduct a self-consistency analysis by rescoring each pilot chain three independent times at temperature 0.5. For each evaluation dimension, self-consistency is computed as the mean pairwise agreement among the three resulting scores.

All five dimensions achieve Krippendorff’s $\alpha \geq 0.85$, exceeding the standard threshold for high reliability (Table 6a, Appendix A.1).

Dimension decoupling. On the 508 complete chains from the original three-condition evaluation (170 datasets evaluated with single-shot, uncompiled-chain, and LIDA baselines; two failed LIDA evaluations excluded), all text–vision correlations are below 0.15, while the Focus–Hypothesis correlation is 0.50 (Figure 4). In the expanded analysis, Focus–Hypothesis correlations range from 0.57 to 0.87 across human and agent annotations, compared with 0.76 for the primary judge. This pattern across datasets and annotators suggests that the correlation partly reflects the pipeline dependency whereby the Test stage builds on the direction established during Focus, while also indicating residual rubric overlap. When we merge FOCUS

and HYPOTHESIS into one dimension with a combined weight of 0.45, the rankings of all nine methods remain unchanged in both settings (Kendall’s $\tau = 1.0$), with VG-COPRO ranked first. We retain the dimensions separately because their responses across controlled quality regimes provide distinct diagnostic information.

Moreover, Cronbach’s α for the five IPM dimensions remains well below the standard 0.7 threshold, indicating that the dimensions do not collapse into a single underlying factor. In particular, removing VISUAL QUALITY is the only modification that increases α , suggesting that the visual dimension captures information not covered by the text-based dimensions. This supports our hypothesis that visualization quality is only weakly correlated with textual reasoning quality, motivating vision-guided evaluation and optimization.

Cross-family robustness. Because the Gemini judge belongs to the same model family as the task model, there is a potential risk of self-enhancement bias, where the judge may overrate generated outputs. To evaluate this possibility, we re-scored 25 chains per condition from three step-matched baselines and five published APO methods using two cross-family evaluators, GPT-4.1 and Qwen-VL-Max. The results show that Gemini assigns lower mean scores than both cross-family models (Table 6b, Appendix A.1), while per-condition Kendall’s τ values remain non-negative across evaluators, suggesting the method comparisons reported in § 5.3 are unlikely to be artifacts of judge-model bias.

5.2 Chain vs. Baselines

We compare InsightChain to step-matched non-chain baselines (Single-shot 4-prompt and LIDA) under two task models on an in-domain test ($n = 35$) and a cross-domain test ($n = 135$).

InsightChain vs. step-matched baselines. InsightChain consistently outperforms the single-shot baseline across different task models (Table 1). The improvement margins on Gemini 3 Flash and GPT-4.1 are highly similar, suggesting that the gains mainly come from the multi-stage pipeline design rather than model-specific behavior. Both models also show a small positive cross-domain transfer gain, indicating that the pipeline generalizes well across domains.

IPM vs Other Visualization Metrics We also re-scored every chain on the cross-domain test set using three published evaluators: CLIP image-caption similarity (Radford et al., 2021), Chart-to-Text BLEU (Kantharaj et al., 2022), and the official LIDA VLM evaluator (Dibia, 2023). Surprisingly, none of them captures the chain’s insight advantage, yet this is understandable since they focus on surface-level alignment (CLIP), caption overlap (BLEU), or code correctness (LIDA), leading to saturation or floor effects and failing to distinguish deeper insights (Table 2a). This directly confirms the motivation for proposing IPM.

Judge-free code validation. We further validate the IPM improvements using two judge-independent code metrics (Table 2b): the frequency of p -value reporting and the number of distinct dataframe variables analyzed. The former is extracted using regex matching over generated code, execution outputs, and narrative text, while the latter measures how broadly the pipeline explores the dataset schema through unique dataframe accesses. Both metrics consistently favor InsightChain over the single-shot baseline ($d \approx 0.4$, $p < 10^{-5}$), with even larger improvements over LIDA. These results suggest that InsightChain performs deeper statistical analysis and broader data exploration. In particular, the dedicated Test stage encourages explicit statistical testing and p -value reporting, while the Explore-Focus workflow promotes wider schema inspection before narrowing to specific hypotheses.

5.3 APO on InsightChain

All five published APO methods fail to improve performance on the cross-domain set, with results falling below the pre-APO baseline and showing no significant gains (Table 1). Among them, only OPRO achieves a positive in-domain improvement. We observe two key patterns. First, under limited optimization budgets, search-based APO methods provide little benefit: the no-search Plan-and-Solve baseline performs comparably to MIPROv2 and COPRO, suggesting that iterative search cannot outperform a strong fixed plan-and-execute scaffold under limited budgets. Second, score-blind proposers such as COPRO and EvoPrompt perform the worst. Although they generate prompts that appear more sophisticated, they often disrupt the underlying reasoning chain instead of improving it.

Appendix A.6 presents checkpoint-reconstructed optimization trajectories. Within the three-iteration

Task model	Method	In-domain (n=35)				Cross-domain (n=135)			
		IPM	Δ	p_1	d	IPM	Δ	p_1	d
Gemini 3 Flash	InsightChain (pre-APO)	7.24	–	–	–	7.38	–	–	–
	Single-shot 4-prompt	6.05	–1.19	1.7×10^{-7}	1.07	6.38	–1.00	1.6×10^{-22}	1.01
	LIDA	5.79	–1.46	5.6×10^{-10}	1.40	5.93	–1.45	1.0×10^{-23}	1.05
	MIPROv2	7.20	–0.05	0.63	–	7.15	–0.24	1.00	–
	COPRO	7.08	–0.16	0.82	–	7.05	–0.33	1.00	–
	OPRO	7.55	+0.30	0.085	0.24	7.27	–0.11	0.86	–
	EvoPrompt	6.88	–0.37	0.96	–	7.00	–0.39	1.00	–
	Plan-and-Solve	7.07	–0.17	0.90	–	7.08	–0.31	1.00	–
GPT-4.1	VG-COPRO	8.56	+1.31	$< 10^{-5}$	1.41	8.54	+1.16	$< 10^{-5}$	1.06
	InsightChain (pre-APO)	7.90	–	–	–	8.01	–	–	–
	Single-shot 4-prompt	6.72	–1.18	7.0×10^{-6}	0.86	6.92	–1.09	3.7×10^{-18}	0.86
	LIDA	4.29	–3.62	9.4×10^{-16}	2.40	4.33	–3.69	2.5×10^{-63}	2.80

Table 1: Main-result table. Δ = method – pre-APO at the same task model; p_1 is a one-sided paired t -test; d is the paired Cohen’s effect size. Negative Δ means the pre-APO wins. Within each task model, rows are grouped as: (a) InsightChain before APO optimization, (b) non-chain baselines, and (c) InsightChain with APO.

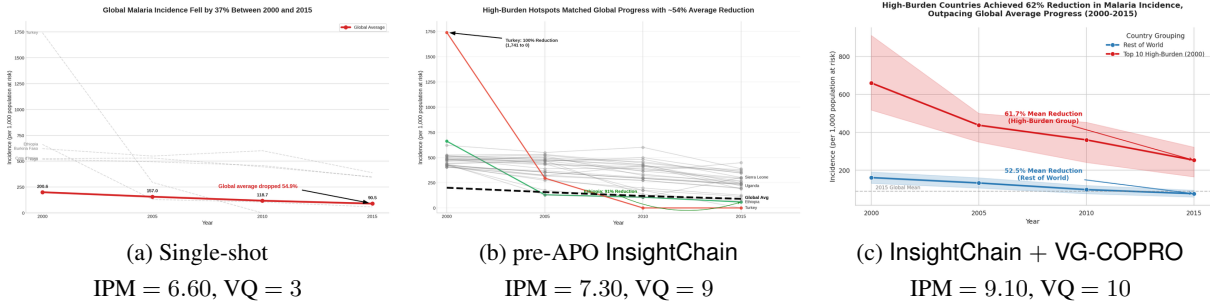


Figure 3: Qualitative comparison of visualizations on the MALARIA_INC dataset. (a) Non-chain baseline: Produces a misleading chart where the title (37%) contradicts the annotated value (54.9%), triggering faithfulness-gate violation (VQ capped at 3). (b) InsightChain pipeline (pre-APO): Correct analytical pattern but poor layout and visual clutter. (c) InsightChain with VG-COPRO: A dual-trend plot with clear annotations and insights.

budget, these trajectories show that optimization can improve validation performance, but that the gains are stage- and method-dependent within our three-iteration budget.

These results reveal a key limitation of existing APO methods: they optimize solely based on textual artifacts from the reasoning chain. As a result, the optimized prompts often improve text-based metrics while failing to generate visually faithful or well-annotated figures. Overall, this suggests that the optimization signal should match the multimodal nature of the final evaluation, motivating the design of VG-COPRO.

5.4 VG-COPRO on InsightChain

VG-COPRO improves IPM over the pre-APO baseline on both splits, with $p_1 < 10^{-5}$ and $d > 1$, and achieves the highest effect size among all APO methods examined, including the five APO methods (Table 1) and the three single-module ablations

(Table 3). It also surpasses the OPRO baseline by more than one IPM point on both splits. Figure 3 illustrates this progression qualitatively on the cross-domain dataset MALARIA_INC; Appendix A.7 provides three additional stage-by-stage traces with dataset identifiers, statistical decisions, final figures, and per-dimension IPM scores.

To assess run-to-run stability, we evaluate three independent VG-COPRO compilations initialized with different seeds. Cross-domain improvements over the pre-APO chain are +1.16, +0.44, and +0.91 IPM points (mean $\pm$ standard deviation: $+0.84 \pm 0.37$), and all are statistically significant ($p \leq 0.02$). In-domain improvements are likewise positive at +1.31, +0.77, and +0.49 (mean $\pm$ standard deviation: $+0.86 \pm 0.42$). Thus, although the magnitude varies across runs, the direction of improvement remains stable.

VG-COPRO’s execution log provides stage-level diagnostic evidence. Under Mod 1, VISUAL QUAL-

	Chain Single LIDA			Δ	d
(a) Other Evaluation Metrics ($n = 135$)					
CLIP img-cap.	0.305	0.300	0.300	+0.00	0.05
C2T BLEU	~ 0	~ 0	~ 0	—	—
LIDA VLM	9.38	9.47	9.38	−0.10	−0.19
IPM (ours)	7.36	6.32	5.97	+1.04	1.07
(b) Judge-free Code Validation ($n = 133$)					
p -value reports	1.13	0.26	0.20	+0.87	0.43
df vars	6.19	5.19	3.47	+1.00	0.42

Table 2: Results on cross-domain test set (means reported for InsightChain, Single-shot, and LIDA). Δ and d measure the difference and effect size between InsightChain and Single-shot. **(a)** Three common visualization metrics fail to separate the methods, while IPM distinctly separates them ($d > 1$). **(b)** Two code metrics favor InsightChain ($p < 10^{-5}$).

ITY is the weakest dimension for the Focus, Test, and Present modules, while HYPOTHESIS RIGOR is weakest for Explore. Under Mod 2, the SPORTS fold is most often the lowest-scoring LODO fold for Explore and Test and ties with CLIMATE for Focus; CLIMATE is most often weakest for Present.

These diagnostics steer the proposer toward prompts that prioritize insight-bearing annotations and appropriate visual encodings, rather than implementation-level Matplotlib detail that can improve text-judged artifacts without improving the rendered figure.

Ablation. We conduct an ablation experiment to attribute the contribution of each modification. As shown in Table 3, each modification provides a significant improvement on both in-domain and cross-domain splits, so the three modifications are not redundant. Mod 2 (LODO) is the strongest single modification on mean Δ , while the full combination yields the highest paired Cohen’s d on both splits, suggesting the effectiveness of our proposed three modifications in VG-COPRO.

5.5 Generalization to Text-to-Image Models

A separate generalization question is whether the chain advantage persists when the Present stage is rendered by a text-to-image (T2I) model rather than a code model. To test this, we replace the Present-stage task model with a T2I variant. The chain outputs a 200–400-word visual brief which gemini-3.1-flash-image-preview renders directly, while the three reasoning stages retain their VG-COPRO-compiled instructions. The step-matched single-shot T2I baseline maintains the same four-prompt single-shot architecture as in

Variant	In-dom ($n=35$)			Cross-dom ($n=135$)		
	IPM	Δ	d	IPM	Δ	d
Hybrid	7.55	+0.49 [†]	0.28	7.27	+0.24 [†]	0.19
+ Mod 1	8.54	+1.30	1.06	8.65	+1.26	0.84
+ Mod 2	8.82	+1.57	1.47	8.74	+1.36	1.06
+ Mod 3	8.50	+1.26	0.93	8.61	+1.23	0.95
VG-COPRO	8.74	+1.49	1.62	8.63	+1.24	1.15

Table 3: Ablation of the three modifications for VG-COPRO (§ 3.4). Hybrid baseline is based on the results of OPRO on chain pipeline. Each modification produces a significant improvement on both splits (all $p_1 < 10^{-5}$ for rows other than the Hybrid baseline). Mod 2 yields the largest mean Δ ; the full combination yields the largest paired Cohen’s d , indicating lowest dispersion.

§ 5.2, with each prompt using the T2I Present-stage task model. Results are summarised in Table 4. The performance improvement (Δ) reported in Table 4 is similar to that observed in Table 1. This suggests that the benefits of InsightChain mainly come from its multi-stage chain-and-feedback workflow itself, rather than from a specific rendering backend such as code-based plotting or T2I generation.

Split	n	Chain	Single-shot	Δ	d
In-domain	35	8.04	5.45	+2.59	2.17
Cross-domain	135	7.60	5.81	+1.79	1.50

Table 4: T2I transferability: mean IPM of InsightChain (T2I Present) vs. single-shot T2I baseline on the full test set. n refers to the number of sampled datasets. All $p_1 < 10^{-14}$ by one-sided paired t -test.

6 Conclusion

We introduce InsightChain, an execution-grounded four-stage pipeline for open-ended insight discovery and visualization, together with IPM and the vision-guided optimizer VG-COPRO. The main experiments show that IPM separates controlled quality levels and agrees with human judgments, while InsightChain outperforms the included step-matched baselines across task models, with judge-free indicators supporting deeper analysis. Existing APO methods provide no consistent gain across both evaluation splits; VG-COPRO improves the chain in both settings, and the optimized pipeline retains its advantage with text-to-image rendering.

Limitations

IPM construct validity. Full five-dimensional human ratings are limited to the 100-chain pilot.

The expanded 300-chain evaluation relies on repeated agent-based ratings and one expert’s audit of three binary criteria. This evidence supports metric repeatability and separation of broad quality regimes, but does not fully validate every individual dimension or IPM’s ability to capture fine-grained, context-dependent insight quality.

Baseline implementation fidelity. The Gemini-row single-shot and LIDA-style baselines are in-house DSPy reproductions of those systems’ architectural patterns rather than direct invocations of the published packages; this holds the task model, judge, and sandbox constant at the cost of head-to-head fidelity. The GPT-4.1 row instead invokes the installed lida v0.0.14 package directly and finds an even larger chain advantage, so the Gemini-row LIDA is the more conservative of the two comparisons. The evaluation also lacks a representative general-purpose data-analysis agent adapted with a visualization stage. Consequently, the results establish an advantage over the included executable visualization and prompting baselines, but not over the broader class of agentic exploratory-data-analysis systems.

Prompt-level overfitting. The five training domains, 20-task validation pool, and three independent compilation runs limit the diversity of optimization conditions. Positive gains across 135 tasks from five unseen domains and across all three runs reduce concerns about test leakage and run-to-run instability, but do not fully rule out prompt-level overfitting.

References

- Nan Chen, Yuge Zhang, Jiahang Xu, Kan Ren, and Yuqing Yang. 2025. [VisEval: A benchmark for data visualization in the era of large language models](#). *IEEE Transactions on Visualization and Computer Graphics*, 31(1):1301–1311.
- Yiming Chen, Chen Zhang, Danqing Luo, Luis Fernando D’Haro, Robby T. Tan, and Haizhou Li. 2024. [Unveiling the achilles’ heel of NLG evaluators: A unified adversarial framework driven by large language models](#). In *Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and virtual meeting, August 11-16, 2024*, volume ACL 2024 of *Findings of ACL*, pages 1359–1375. Association for Computational Linguistics.
- Liyang Cheng, Xingxuan Li, and Lidong Bing. 2023. [Is gpt-4 a good data analyst?](#) *Preprint*, arXiv:2305.15038.
- Victor Dibia. 2023. [LIDA: A tool for automatic generation of grammar-agnostic visualizations and infographics using large language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 113–126, Toronto, Canada. Association for Computational Linguistics.
- Rui Ding, Shi Han, Yong Xu, Haidong Zhang, and Dongmei Zhang. 2019. [QuickInsights: Quick and automatic discovery of insights from multi-dimensional data](#). In *Proceedings of the 2019 International Conference on Management of Data (SIGMOD)*, pages 317–332.
- Kuicai Dong, Shurui Huang, Fangda Ye, Wei Han, Zhi Zhang, Dexun Li, Wenjun Li, Qu Yang, Gang Wang, Yichao Wang, Chen Zhang, and Yong Liu. 2026. [Doc-researcher: A unified system for multimodal document parsing and deep research](#). In *Proceedings of the ACM Web Conference 2026, WWW 2026*, pages 2349–2360. ACM.
- Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. 2024. [Connecting large language models with evolutionary algorithms yields powerful prompt optimizers](#). In *The Twelfth International Conference on Learning Representations*.
- Jeffrey Heer and Ben Shneiderman. 2012. [Interactive dynamics for visual analysis](#). *Communications of the ACM*, 55:45–54.
- Sirui Hong, Yizhang Lin, Bang Liu, Bangbang Liu, Binhao Wu, Ceyao Zhang, Danyang Li, Jiaqi Chen, Jiayi Zhang, Jinlin Wang, Li Zhang, Lingyao Zhang, Min Yang, Mingchen Zhuge, Taicheng Guo, Tuo Zhou, Wei Tao, Robert Tang, Xiangtao Lu, and 9 others. 2025. [Data interpreter: An LLM agent for data science](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 19796–19821. Association for Computational Linguistics.
- Shankar Kantharaj, Rixie Tiffany Leong, Xiang Lin, Ahmed Masry, Megh Thakkar, Enamul Hoque, and Shafiq Joty. 2022. [Chart-to-text: A large-scale benchmark for chart summarization](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4005–4023, Dublin, Ireland. Association for Computational Linguistics.
- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan A, Saiful Haq, Ashutosh Sharma, Thomas Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. 2024. [Dspy: Compiling declarative language model calls into state-of-the-art pipelines](#). In *International Conference on Learning Representations*, volume 2024, pages 54928–54958.
- Fangyu Lei, Jinxiang Meng, Yiming Huang, Junjie Zhao, Yitong Zhang, Jianwen Luo, Xin Zou, Ruiyi Yang, Wenbo Shi, Yan Gao, Shizhu He, Jun Zhao, Zuo

- Wang, Qian Liu, Yang Wang, WANG KE, and Kang Liu. 2026. [DAComp: Benchmarking data agents across the full data intelligence lifecycle](#). In *The Fourteenth International Conference on Learning Representations*.
- Shuaimin Li, Xuanang Chen, Yuanfeng Song, Yunze Song, Chen Jason Zhang, Fei Hao, and Lei Chen. 2025a. [prompt4vis: prompting large language models with example mining for tabular data visualization](#). *The VLDB Journal*, 34(4):38.
- Wenjun Li, Zhi Chen, Jingru Lin, Hannan Cao, Wei Han, Sheng Liang, Zhi Zhang, Kuicai Dong, Dexun Li, Chen Zhang, and Yong Liu. 2025b. [Reinforcement learning foundations for deep research systems: A survey](#). *CoRR*, abs/2509.06733.
- Sheng Liang, Yongyue Zhang, Nathanael Brian, Hang Lv, Hao Wang, Chen Zhang, and Yong Liu. 2026. [AsymSpec: Context-asymmetric speculative decoding for agentic LLMs](#). *Preprint*, arXiv:2608.26004.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. [G-Eval: NLG evaluation using GPT-4 with better human alignment](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522.
- Yuyu Luo, Nan Tang, Guoliang Li, Chengliang Chai, Wenbo Li, and Xuedi Qin. 2021. [Synthesizing natural language to visualization \(NL2VIS\) benchmarks from NL2SQL benchmarks](#). In *Proceedings of the 2021 International Conference on Management of Data*, pages 1235–1247. Association for Computing Machinery.
- Hang Lv, Sheng Liang, Hongchao Gu, Wei Guo, Defu Lian, Yong Li, Hao Wang, and Enhong Chen. 2026a. [IE as cache: Information extraction enhanced agentic reasoning](#). *CoRR*, abs/2604.14930.
- Hang Lv, Sheng Liang, Hao Wang, Hongchao Gu, Yaxiong Wu, Wei Guo, Defu Lian, Yong Liu, and Enhong Chen. 2025. [Costeer: Collaborative decoding-time personalization via local delta steering](#). *CoRR*, abs/2507.04756.
- Hang Lv, Sheng Liang, Hao Wang, Yongyue Zhang, Hongchao Gu, Wei Guo, Defu Lian, Yong Liu, and Enhong Chen. 2026b. [Specsteer: Synergizing local context and global reasoning for efficient personalized generation](#). *CoRR*, abs/2603.16219.
- Pingchuan Ma, Rui Ding, Shi Han, and Dongmei Zhang. 2021. [MetaInsight: Automatic discovery of structured knowledge for exploratory data analysis](#). In *Proceedings of the 2021 International Conference on Management of Data (SIGMOD)*, pages 1262–1274.
- Pingchuan Ma, Rui Ding, Shuai Wang, Shi Han, and Dongmei Zhang. 2023. [InsightPilot: An LLM-empowered automated data exploration system](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 346–352.
- Paula Maddigan and Teo Susnjak. 2023. [Chat2VIS: Generating data visualizations via natural language using ChatGPT, codex and GPT-3 large language models](#). *IEEE Access*, 11:45181–45193.
- Bodhisattwa Prasad Majumder, Harshit Surana, Dhruv Agarwal, Bhavana Dalvi Mishra, Abhijeetsingh Meena, Aryan Prakhar, Tirth Vora, Tushar Khot, Ashish Sabharwal, and Peter Clark. 2025. [Discoverybench: Towards data-driven discovery with large language models](#). In *The Thirteenth International Conference on Learning Representations*.
- Tamara Munzner. 2014. [Visualization Analysis and Design](#). CRC Press, Boca Raton, Florida.
- Krista Opsahl-Ong, Michael J Ryan, Josh Purtell, David Broman, Christopher Potts, Matei Zaharia, and Omar Khattab. 2024. [Optimizing instructions and demonstrations for multi-stage language model programs](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 9340–9366, Miami, Florida, USA. Association for Computational Linguistics.
- Peter Pirolli and Stuart Card. 2005. [The sensemaking process and leverage points for analyst technology as identified through cognitive task analysis](#). In *Proceedings of international conference on intelligence analysis*, volume 5, pages 2–4. McLean, VA, USA.
- Reid Pryzant, Dan Iter, Jerry Li, Yin Tat Lee, Chenguang Zhu, and Michael Zeng. 2023. [Automatic prompt optimization with “gradient descent” and beam search](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7957–7968.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. [Learning transferable visual models from natural language supervision](#). In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR.
- Dominik Sacha, Andreas Stoffel, Florian Stoffel, Bum Chul Kwon, Geoffrey Ellis, and Daniel A. Keim. 2014. [Knowledge generation model for visual analytics](#). *IEEE Transactions on Visualization and Computer Graphics*, 20(12):1604–1613.
- Shiori Sagawa*, Pang Wei Koh*, Tatsunori B. Hashimoto, and Percy Liang. 2020. [Distributionally robust neural networks](#). In *International Conference on Learning Representations*.
- Gaurav Sahu, Abhay Puri, Juan A. Rodriguez, Amirhossein Abaskohi, Mohammad Chegini, Alexandre Drouin, Perouz Taslakian, Valentina Zantedeschi, Alexandre Lacoste, David Vazquez, Nicolas Chapados, Christopher Pal, Sai Rajeswar, and Issam H. Laradji. 2025. [Insightbench: Evaluating business](#)

- analytics agents through multi-step insight generation. In *The Thirteenth International Conference on Learning Representations*.
- Danqing Shi, Xinyue Xu, Fuling Sun, Yang Shi, and Nan Cao. 2021. [Calliope: Automatic visual data story generation from a spreadsheet](#). *IEEE Transactions on Visualization and Computer Graphics*, 27(2):453–463.
- Yuan Tian, Weiwei Cui, Dazhen Deng, Xinjing Yi, Yurun Yang, Haidong Zhang, and Yingcai Wu. 2025. [Chartgpt: Leveraging llms to generate charts from abstract natural language](#). *IEEE Transactions on Visualization and Computer Graphics*, 31(3):1731–1745.
- John W. Tukey. 1977. *Exploratory Data Analysis*. Addison-Wesley, Reading, Massachusetts.
- Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. 2023. [Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2609–2634, Toronto, Canada. Association for Computational Linguistics.
- Yun Wang, Zhida Sun, Haidong Zhang, Weiwei Cui, Ke Xu, Xiaojuan Ma, and Dongmei Zhang. 2020. [DataShot: Automatic generation of fact sheets from tabular data](#). *IEEE Transactions on Visualization and Computer Graphics*, 26(1):895–905.
- Yaxiong Wu, Sheng Liang, Chen Zhang, Yichao Wang, Yongyue Zhang, Huifeng Guo, Ruiming Tang, and Yong Liu. 2025. [From human memory to AI memory: A survey on memory mechanisms in the era of llms](#). *CoRR*, abs/2504.15965.
- Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V Le, Denny Zhou, and Xinyun Chen. 2024. [Large language models as optimizers](#). In *The Twelfth International Conference on Learning Representations*.
- Zhaorui Yang, Bo Pan, Han Wang, Yiyao Wang, Xingyu Liu, Luoxuan Weng, Yingchaojie Feng, Haozhe Feng, Minfeng Zhu, Bo Zhang, and Wei Chen. 2026. [Multimodal deepresearcher: Generating text-chart interleaved reports from scratch with agentic framework](#). In *Fortieth AAAI Conference on Artificial Intelligence, Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence, Sixteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2026, Singapore, January 20-27, 2026*, pages 34368–34377. AAAI Press.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. [ReAct: Synergizing reasoning and acting in language models](#). In *International Conference on Learning Representations (ICLR)*.
- Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin, and James Zou. 2025. [Optimizing generative AI by backpropagating language model feedback](#). *Nature*, 639(8055):609–616.
- Chen Zhang, Dading Chong, Feng Jiang, Chengguang Tang, Anningzhe Gao, Guohua Tang, and Haizhou Li. 2025. [Aligning language models using follow-up likelihood as reward signal](#). In *Thirty-Ninth AAAI Conference on Artificial Intelligence, Thirty-Seventh Conference on Innovative Applications of Artificial Intelligence, Fifteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2025, Philadelphia, PA, USA, February 25 - March 4, 2025*, pages 25832–25841. AAAI Press.
- Chen Zhang, Luis Fernando D’Haro, Yiming Chen, Malu Zhang, and Haizhou Li. 2024a. [A comprehensive analysis of the effectiveness of large language models as automatic dialogue evaluators](#). In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20-27, 2024, Vancouver, Canada*, pages 19515–19524. AAAI Press.
- Chen Zhang, Kuicai Dong, Dexun Li, Wenjun Li, Qu Yang, Wei Han, and Yong Liu. 2026. [Srr-judge: Step-level rating and refinement for enhancing search-integrated reasoning in search agents](#). *CoRR*, abs/2602.07773.
- Chen Zhang, Chengguang Tang, Dading Chong, Ke Shi, Guohua Tang, Feng Jiang, and Haizhou Li. 2024b. [Ts-align: A teacher-student collaborative framework for scalable iterative finetuning of large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, Florida, USA, November 12-16, 2024*, volume EMNLP 2024 of *Findings of ACL*, pages 8926–8946. Association for Computational Linguistics.
- Ran Zhang and Mohannad Elhamod. 2025. [Data-to-dashboard: Multi-agent llm framework for insightful visualization in enterprise analytics](#). *Preprint*, arXiv:2505.23695.
- Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. 2023. [Large language models are human-level prompt engineers](#). In *International Conference on Learning Representations (ICLR)*.
- Zhenghao Zhu, Yuanfeng Song, Xing Chen, Chengzhong Liu, Yakun Cui, Caleb Chen Cao, Sirui Han, and Yike Guo. 2026. [InsightEval: An expert-curated benchmark for assessing insight discovery in LLM-driven data agents](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 34632–34656. Association for Computational Linguistics.

A Supplementary Material

A.1 IPM Validation Tables

This subsection presents the supporting results for § 5.1, including per-dimension human–IPM agreement (Table 5), judge self-consistency and cross-family validation (Table 6), and the inter-dimension correlation heatmap (Figure 4).

Dimension	r	ρ	MAE	$\mathbf{h} - \mathbf{IPM}$
Exploration Breadth	0.78	0.76	2.53	+2.18
Focus Relevance	0.75	0.75	1.54	+1.17
Hypothesis Rigor	0.53	0.51	2.63	+2.24
Insight Communication	0.65	0.67	2.53	+2.22
Visual Quality	0.68	0.77	1.79	−0.04
Composite	0.87	0.88	1.88	+1.86

Table 5: Human–IPM agreement on $n = 97$ blind-annotated pilot chains. r , ρ = Pearson / Spearman correlations between human and IPM scores on the 1–10 scale (all $p < 10^{-7}$); MAE = mean absolute error; $\mathbf{h} - \mathbf{IPM}$ = human-mean minus IPM-mean (the scale-offset bias). Composite row is human overall-quality vs. the IPM 5-dim weighted composite. The vision-judged VISUAL QUALITY is essentially calibrated to the human ($\Delta = -0.04$); the four text dimensions show a uniform +1.2-to-+2.2 strictness offset that does not affect rank-based agreement.

A.2 IPM Expanded Validation and Close-Pair Evaluation

Expanded agent-based evaluation. Because the 100-chain human study is limited in scale, we expanded the evaluation to 300 generated analysis chains spanning all ten domains. The 300 chains consist of the 100-chain pilot (20 datasets under five regimes) with 200 additional chains generated from 50 datasets under four regimes. The resulting condition counts are 70 HIGH, 90 MEDIUM, 70 LOW, and 70 RANDOM. Each annotation record contains the four stage outputs, generated code, execution output, final insight, and rendered Present-stage figure. Annotators assign five integer dimension scores on the 1–10 scale, an independent overall-quality score, and categorical judgments of factual correctness, practical meaningfulness, and effect-size reporting.

An advanced proprietary agent system independently rated every chain three times using the same rubric provided to the human annotators. Its mean ratings show strong agreement with IPM (Pearson $r = 0.88$, Spearman $\rho = 0.89$, and MAE = 0.93), while the repeated ratings show high test-

(a) Per-dimension self-consistency (3 runs)			
Dimension	n	α	$\bar{r}$
Exploration Breadth	100	0.959	0.959
Focus Relevance	100	0.894	0.894
Hypothesis Rigor	100	0.918	0.918
Insight Communication	100	0.934	0.935
Visual Quality	99	0.926	0.928

(b) Cross-family validation vs. in-family Gemini			
Judge	n	Gem / XJ	Δ
GPT-4.1	200	6.95 / 6.99	+0.04
Qwen-VL-Max	199	6.89 / 7.57	+0.68

Table 6: IPM judge reliability. (a) Per-dimension self-consistency on the 100-chain pilot: all five dimensions are independently rescored three times at temperature 0.5; n = number of evaluated chains with complete three-run scores, α = Krippendorff’s interval alpha across three judge runs, and $\bar{r}$ = mean pairwise Pearson r . (b) Cross-family validation against the in-family Gemini 3 Flash judge: “Gem” is the in-family mean on the matched subsample, “XJ” is the cross-family judge’s mean, and $\Delta = \text{XJ} - \text{Gem}$ reports the mean score difference. Both cross-family judges return means at or above Gemini, providing no evidence of in-family score inflation on this matched subsample.

retest reliability (overall Krippendorff’s $\alpha = 0.88$; per-dimension $\alpha = 0.63$ – 0.91). A data-analysis expert then audited the agent’s judgments of factual correctness, effect-size reporting, and practical meaningfulness, correcting incorrect or ambiguous binary decisions. Based on these expert-corrected labels, practical meaningfulness rises from 0% for RANDOM/LOW outputs to 95.4% for HIGH outputs. These results support the repeatability of IPM and its separation of broad quality regimes, but do not by themselves establish the construct validity of every individual dimension.

Close-pair evaluation. To test whether IPM distinguishes outputs of similar quality, agent annotators compared 80 pairs, including regenerated outputs from the same quality regime and outputs with nearly identical IPM scores. The 80 pairs are drawn from three sources:

Source	Pairs	Construction
Seed pair	20	Same dataset and prompt; different seed
Regeneration	40	Same dataset and quality regime
Metric-close	20	Same regime, different datasets, $ \Delta \text{IPM} < 0.5$

Table 7: Composition of the close-pair evaluation set.

Pairs without a rendered Present-stage figure

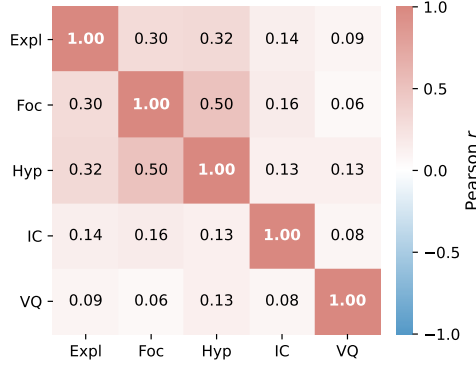


Figure 4: Inter-dimension Pearson correlations on 508 complete chains from the original three-condition evaluation: 170 datasets evaluated with single-shot, uncompiled-chain, and LIDA baselines, excluding two failed LIDA evaluations. Mean off-diagonal $\bar{r} = 0.19$; max $r = 0.50$ (Foc–Hyp). Cronbach’s α on the full 5-tuple is 0.39; the leave-one-out values are –Expl 0.37, –Foc 0.29, –Hyp 0.22, –IC 0.37, –VQ 0.44 — VISUAL QUALITY is the only dimension whose removal raises α , consistent with its capturing variance not covered by the four text-judged dimensions. Expl = Exploration Breadth, Foc = Focus Relevance, Hyp = Hypothesis Rigor, IC = Insight Communication, VQ = Visual Quality.

are excluded. Pair order and left–right assignment are randomized, while the quality regime and IPM scores remain hidden during annotation. The blind interface presents the two figures and final insights and requests an A, B, or tie judgment. Preferences are compared with the sign and magnitude of the hidden IPM difference only after annotation. Based on this analysis, differences below approximately 0.8 IPM points are interpreted as ties. The results indicate that IPM reliably distinguishes outputs when $|\Delta\text{IPM}| > 0.8$, whereas smaller differences behave as ties.

Blinding and repeated agent ratings. Before annotation, condition labels, automatic IPM scores, tier-bearing chain identifiers, and identifying figure paths are removed or replaced with neutral identifiers. The private identifier mapping is not reopened until a complete annotation pass has been frozen. The first two passes divide the corpus into 20 blinded packets of 15 chains, with independently scored packets and different deterministic orderings; the third pass uses 300 individually masked packets and neutralized figure names. All three passes use the same rubric as the human pilot. For agreement with IPM, the three-run mean for each dimension is combined using the original

IPM weights and compared with the primary IPM composite. Test–retest reliability is computed as interval Krippendorff’s α across the three passes.

Expert audit. The binary criteria are audited against execution-grounded evidence. For factual correctness, the auditor checks every reported number and direction against the chain’s executed stdout and, where relevant, the rendered figure. Effect-size reporting requires an explicit magnitude, such as a difference, percentage, ratio, correlation, or standardized effect, rather than a p -value alone. Practical meaningfulness is judged independently of factual correctness and distinguishes domain-relevant findings from trivial descriptions. The audit artifacts contain 37 deliberately enriched disagreement cases: 7 for meaningfulness, 15 for effect-size reporting, and 15 for factual correctness. The auditor records an evidence basis, rationale, and corrected verdict for each case. Because this set concentrates on disputed cases, its annotator-agreement rates are not interpreted as population estimates.

A.3 Metric Comparison

Table 8 compares IPM with representative metrics from the three families reviewed in § 2.3 across five design properties.

A.4 APO Design-Space Summary

Table 9 consolidates the five APO methods of § 3.3 along their two distinguishing axes (search algorithm and search target).

A.5 Stage Instructions and VG-COPRO Algorithm

Stage instructions θ_t . The four seed instructions that VG-COPRO mutates (taken verbatim from the DSPy signature docstrings; Explore is also shown in § 3.1):

- **Explore (θ_1):** “Given a dataset summary and sample, generate Python visualization code that broadly explores the data landscape with 2–3 overview charts. Identify distributions, relationships, temporal patterns, and outliers. Print key observations and identify non-obvious patterns worth investigating.”
- **Focus (θ_2):** “Given exploration insights and key patterns, drill down into the most promising pattern. Use groupby, filtering, or segmentation to investigate in depth. Formulate a specific, testable hypothesis.”

Metric	Multi-stage	Vision-judged	Faithfulness gate	Insight-aware	Spec-free
nvBench-style spec match (Luo et al., 2021)	✗	✗	✗	✗	✗
Chart-to-Text BLEU (Kantharaj et al., 2022)	✗	✗	✗	✗	✓
CLIP (Radford et al., 2021)	✗	✓	✗	✗	✓
LIDA partial VLM eval (Dibia, 2023)	✗	partial	✗	partial	✓
InsightBench rubric (Sahu et al., 2025)	text only	✗	✗	✓	✓
IPM (ours)	✓	✓	✓	✓	✓

Table 8: Visualization quality metrics by design property. **Multi-stage**: does the metric score intermediate chain stages rather than only the final figure? **Vision-judged**: does it consume the rendered image? **Faithfulness gate**: is there an explicit cap for misleading figures? **Insight-aware**: does it reward task-specific finding extraction? **Spec-free**: does it avoid requiring a gold chart specification?

Method	Search algorithm	What is searched
MIPROv2	Bayesian / TPE	instr. + demos
COPRO	Coordinate ascent	instr. only
OPRO	LLM-as-optimizer	instr. only
EvoPrompt	Evolutionary	instr. only
Plan-and-Solve	none (template)	—

Table 9: The five APO paradigms compiled and evaluated under identical splits, metric, and budget. Plan-and-Solve has no learnable parameters and serves as a structured-template control.

- **Test** (θ_3): “Given a hypothesis from the Focus step, design and execute a statistical test to validate or refute it. Use `scipy.stats` for appropriate tests. Report *p*-value, effect size, and confidence intervals.”
- **Present** (θ_4): “Given all prior insights from a full analysis chain, create ONE polished, publication-ready annotated visualization that communicates the key finding. Use annotations, arrows, color, and a headline title that states the finding.”

Algorithm. The full pseudocode for VG-COPRO is given in Algorithm 1. Inline comments indicate where each modification (Mod 1–3) alters the COPRO + OPRO-proposer baseline in § 3.3.

A.6 Optimization Dynamics

Figure 5 reconstructs the incumbent validation trajectory from the released per-method state checkpoints. Iteration 0 is the seed instruction, and iterations 1–3 are the three candidate-proposal rounds. At each round, the plotted point is the validation-pool mean of the incumbent selected according to that method’s own objective. OPRO improves in all four stages, with final stage-wise gains of 0.25, 0.13, 0.14, and 0.30 validation-score points for Explore, Focus, Test, and Present, respectively. Evo-

Algorithm 1 VG-COPRO

Require: chain $\Pi = (\pi_1, \dots, \pi_4)$; LODO folds $\{\mathcal{F}_d\}_{d=1}^5$; proposer g ; text/vision judges J_t, J_v ; iterations $N=3$, history $K=5$

- 1: **for** $\pi \in \Pi$ **do**
- 2: $\mathcal{H} \leftarrow [(\pi, \text{eval}(\Pi))]$
- 3: **for** $t = 1, \dots, N$ **do**
- 4: $\triangleright$ Propose (Mod 1: diagnostic, per-dimension feedback)
- 5: $\pi' \leftarrow g(\text{top-}K(\mathcal{H}))$; replace π with π' in Π
- 6: $\triangleright$ Evaluate (Mod 2: LODO folds; Mod 3: vision-aware IPM)
- 7: **for** $d = 1, \dots, 5$ **do**
- 8: $\bar{s}_d \leftarrow \text{mean}_{x \in \mathcal{F}_d} \text{IPM}(J_t(\Pi(x)) \oplus J_v(\Pi(x).I))$
- 9: **end for**
- 10: $\triangleright$ Aggregate (worst-fold objective)
- 11: $\text{fit} \leftarrow \min_d \bar{s}_d$
- 12: append (π', fit) to $\mathcal{H}$
- 13: **end for**
- 14: $\pi \leftarrow \arg \max_{\pi' \in \mathcal{H}} \text{fit}$
- 15: **end for**

Ensure: compiled chain Π^*

Prompt improves in two stages and is unchanged in the other two. The diagnostic-only and vision-only variants improve in three and two stages, respectively. The frequent plateaus indicate that useful prompt updates are concentrated in particular stages rather than arising uniformly at every iteration. Accordingly, the figure shows that APO can improve stage-level validation scores within individual methods, with gains confined to particular stages and proposal rounds.

These within-objective gains do not necessarily transfer to the multimodal held-out evaluation. The five published APO baselines optimize only textual artifacts from the reasoning chain, whereas the final IPM also evaluates the rendered figure. OPRO provides the clearest example of this mismatch: although its text-only validation trajectory improves in all four stages, its cross-domain full-IPM mean falls from 7.38 before APO to 7.27 after optimiza-

tion (Table 1). Among chains with complete cross-domain dimension ratings, all five published APO methods also score 0.33–0.87 points lower than pre-APO InsightChain on INSIGHT COMMUNICATION and 0.56–1.19 points lower on VISUAL QUALITY. Thus, improvements under a text-only compile objective do not ensure visually faithful, well-annotated figures that communicate the discovered insight. Consistent with this interpretation, the vision-aware Mod 3 uses the full five-dimensional IPM during compilation and improves held-out cross-domain IPM by 1.23 points (Table 3). Together, these results motivate aligning the optimization signal with the multimodal final evaluation, as VG-COPRO does.

A.7 Complete Qualitative Analysis Chains

To make the analytical depth behind the aggregate scores directly inspectable, Figures 6–8 reproduce three VG-COPRO outputs from different domains. The examples cover one in-domain and two held-out cross-domain tasks and include both supported and refuted hypotheses. For readability, we omit the generated Python code but retain the complete semantic progression through Explore, Focus, Test, and Present, including the decision-relevant statistics and caveats. IPM tuples follow the order (EXPLORATION BREADTH, FOCUS RELEVANCE, HYPOTHESIS RIGOR, INSIGHT COMMUNICATION, VISUAL QUALITY).

A.7.1 Healthcare: Mental-Health Survey

Dataset and score. `mental_health_survey` (healthcare; in-domain test; 1,200 rows). IPM tuple = (8, 9, 9, 9, 10); composite IPM = 8.90. Figure 6 shows the final visualization.

Explore. The initial exploration compared stress, productivity, treatment seeking, sleep, and work arrangement. It suggested a weak stress–productivity relationship, higher stress among respondents who sought treatment, and a possible buffering role for sleep, while showing wide productivity variation within every work mode.

Focus. The chain narrowed the question to whether hybrid work changes the stress–productivity relationship. It hypothesized that hybrid workers would have significantly greater productivity variance than in-office workers and that their stress–productivity correlation would be at least 0.10 weaker.

Test. The hypothesis was refuted. Hybrid and in-office productivity standard deviations were 15.44 and 14.65, respectively, but Levene’s test found no significant variance difference ($p = 0.3048$). Their Pearson correlations were $r = 0.094$ and $r = 0.010$, so the hybrid relationship was slightly stronger rather than weaker; Fisher’s z test also found no significant difference ($p = 0.2581$). The chain further noted that all correlations were below $|0.10|$, limiting substantive interpretation.

Present. The final three-panel figure exposes the near-flat regression slopes and comparable dispersions for in-office, hybrid, and remote respondents. Its conclusion is correspondingly negative: work arrangement does not significantly alter either productivity variance or the observed stress–productivity relationship in this sample.

A.7.2 Transportation: Parking Tickets

Dataset and score. `tickets` (transportation; cross-domain test; 50,000 rows). IPM tuple = (8, 9, 10, 9, 10); composite IPM = 9.10. Figure 7 shows the final visualization.

Explore. The chain found that meter-related violations dominate citation volume, ticketing peaks around late morning and early afternoon, and the Philadelphia Parking Authority (PPA) issues nearly 90% of tickets. Police issue fewer citations but span a wider range of fines.

Focus. It connected the agency-level disparity to violation composition: PPA activity concentrates on meter expiration and overtime violations, whereas Police activity includes more vehicle-legality and traffic-safety violations. The resulting hypothesis was that the Police mean fine is at least 25% higher than the PPA mean.

Test. A Welch two-sample t -test supported the hypothesis: Police and PPA means were US\$60.99 and US\$41.81, a 45.87% increase and a US\$19.18 difference ($t = 25.27$, $p < 2.2 \times 10^{-16}$, Cohen’s $d = 0.53$; 95% CI for the difference [US\$17.69, US\$20.67]). The chain cautioned that Police fines include high-value outliers and that the PPA sample is almost ten times larger, while noting that the Police median is also higher (US\$51 vs. US\$36).

Present. The hybrid violin–box plot shows the two full distributions, marks both means, and annotates the 45.9% gap. The final insight attributes the difference to distinct enforcement portfolios

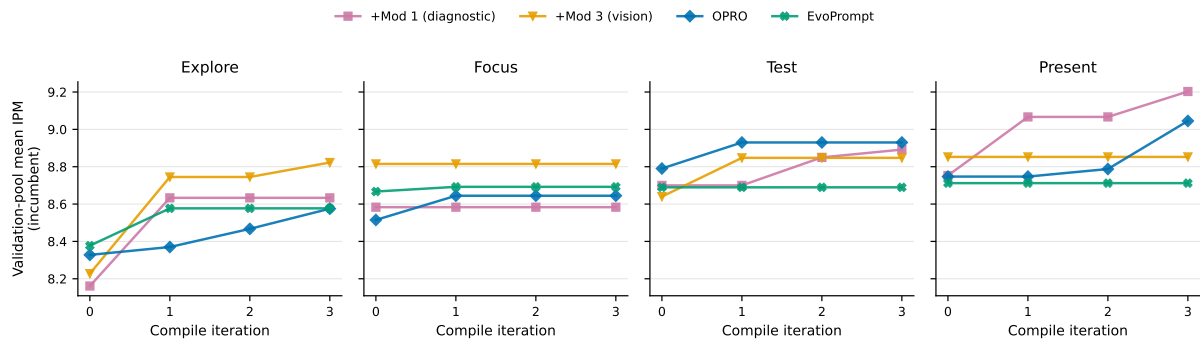


Figure 5: Checkpoint-reconstructed optimization dynamics over three compile iterations. Each panel plots the validation-pool mean of the incumbent selected under each method’s own objective. Curves are comparable over iterations within a method, but absolute heights should not be compared across metric variants: OPRO, EvoPrompt, and Mod 1 use four text dimensions, whereas Mod 3 uses the full five-dimensional IPM. VG-COPRO and the isolated LODO variant are omitted because their worst-fold selection objective is not directly comparable to the plotted mean-score selection trajectories; methods without reconstructable validation-mean histories are also omitted.

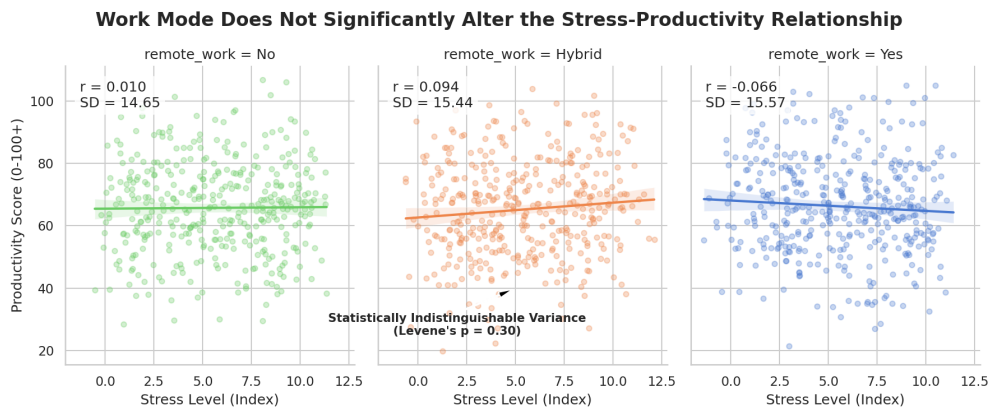


Figure 6: Complete healthcare analysis chain. The example shows the pipeline revising an exploratory impression after formal testing rather than forcing a positive finding.

while presenting the composition explanation as an interpretation rather than a causal estimate.

A.7.3 Ecommerce: Google Play Reviews

Dataset and score. google_play_store_reviews_did43565 (ecommerce; cross-domain test; 14,076 rows). IPM tuple = (9, 9, 9, 10, 10); composite IPM = 9.35. Figure 8 shows the final visualization.

Explore. The chain identified polarized ratings, selective developer replies to low-score reviews, and substantial variation across applications. It proposed examining whether detailed complaints receive more community endorsement through the thumbsUpCount field.

Focus. The drill-down found higher mean engagement for one-star reviews but a median near zero at every score, suggesting that a small number of viral complaints drives the difference. The hy-

pothesis deliberately imposed a strong threshold: one-star reviews would receive at least 200% more mean engagement than five-star reviews.

Test. The thresholded hypothesis was refuted even though the direction was significant. Mean thumbs-up counts were 3.34 for one-star and 1.41 for five-star reviews, a 136.9% increase (Welch’s $t = 4.885$, $p = 1.07 \times 10^{-6}$, Cohen’s $d = 0.134$). The small effect size and a maximum of 397 thumbs-up votes support the chain’s caveat that rare, highly visible complaints drive the mean difference.

Present. A transparent jitter plot retains the dense mass at zero rather than hiding it behind group averages, annotates the two means, and points to the 397-vote outlier. The final insight therefore distinguishes statistically reliable mean separation from the typical review’s near-zero engagement.

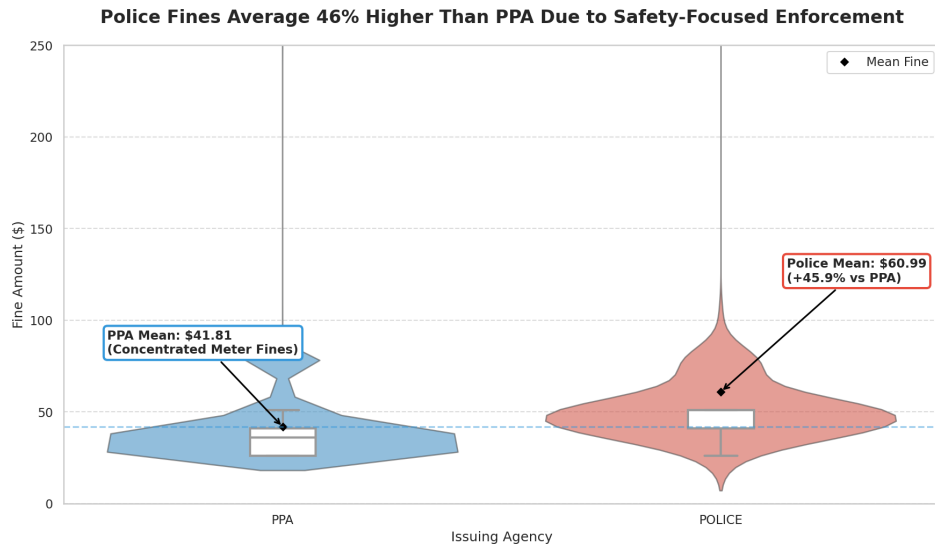


Figure 7: Complete transportation analysis chain. This held-out-domain example progresses from broad enforcement patterns to a quantified agency-level difference with uncertainty and effect size.

A.8 Annotator Handbook

The complete handbook is reproduced in Figures 9–16. Each row is a four-stage analysis chain; dimensions 1–4 are scored from text artifacts, while VISUAL QUALITY is scored from the rendered Present-stage figure. Annotators assign five independent 1–10 dimension scores, a separate overall-quality score, and categorical judgments of factual correctness, meaningfulness, and effect-size reporting. Automatic scores remain hidden until the human judgments are complete.

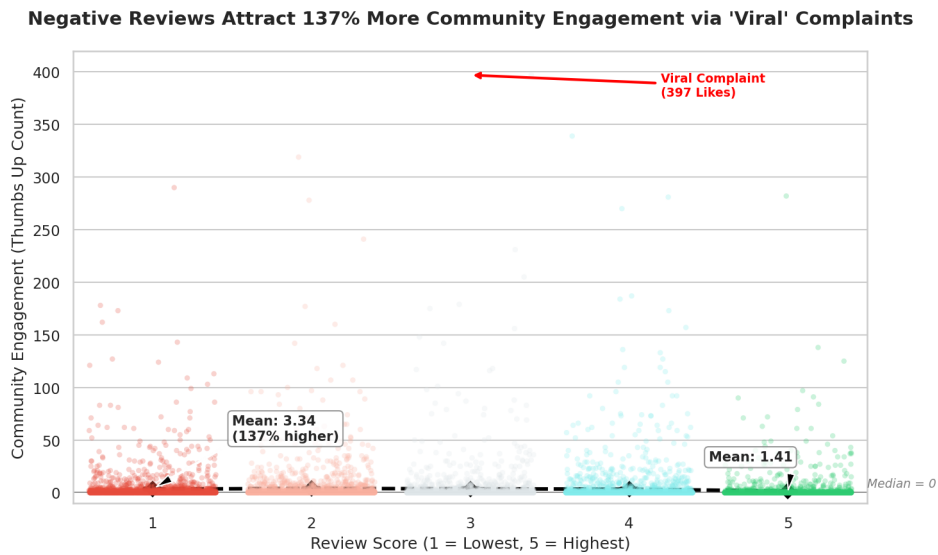


Figure 8: Complete ecommerce analysis chain. This held-out-domain example preserves a nuanced negative decision: the directional effect is significant, but its magnitude does not meet the 200% threshold specified before the Test stage and is dominated by outliers.

Annotation unit and outputs

Each row is one Explore → Focus → Test → Present analysis chain. Dimensions 1–4 use only text artifacts (code, stdout, insights, and narrative). Dimension 5 uses the rendered chart at present figure path; an empty or missing figure scores 1. For every chain, assign five integer scores (1–10), an independent overall-quality score (1–10), three categorical final-insight judgments, and an optional one-sentence note.

1. Exploration Breadth — human_exploration_breadth

How comprehensively does Explore examine the dataset? Read explore_insights and score inquiry scope, not visual appearance.

- 1–3 One aspect only; e.g., one histogram without interpretation.
- 4–6 Two or three aspects, but important characteristics are missed.
- 7–8 Good coverage of distributions, relationships, temporal patterns, or outliers; identifies a non-obvious pattern.
- 9–10 Comprehensive coverage of distributions, relationships, temporal/grouped structure, and outliers; yields multiple deep, non-obvious patterns.

Figure 9: Annotator handbook (1/8): annotation unit, required outputs, and scoring anchors for EXPLORATION BREADTH.

2. Focus Relevance — human_focus_relevance

Does Focus select a relevant, non-obvious direction? Read `focus_insights` and ask whether a thoughtful analyst would choose this direction among the alternatives.

- 1-3 Random or uninteresting pattern.
- 4-6 Reasonable but superficial focus, such as noting different group means.
- 7-8 Meaningful, non-obvious pattern with clear analytical relevance.
- 9-10 The most insightful and actionable pattern in the data.

3. Hypothesis Rigor — human_hypothesis_rigor

How methodologically rigorous is Test? Read `test_insights` and judge scientific appropriateness, not whether the outcome is interesting.

- 1-3 No genuine test; description or unsupported assertion only.
- 4-6 A hypothesis with a weak or inappropriate test (e.g., an unexplained t-test on heavily skewed data).
- 7-8 Clear hypothesis, appropriate test, and clear p-value and/or effect size.
- 9-10 Effect sizes and confidence intervals, with confounders or robustness checks considered.

Figure 10: Annotator handbook (2/8): scoring anchors for FOCUS RELEVANCE and HYPOTHESIS RIGOR.

4. Insight Communication — text only

Field: `human_insight_communication`. Judge whether Present code and narrative communicate a specific, supported insight. Do not inspect the PNG; rendering defects belong to Visual Quality.

- 1-3 Raw chart without annotation calls (`plt.annotate`, `plt.text`, reference lines), or a generic `final_insight` without values, groups, or thresholds.
- 4-6 Clean code, title, or axis labels but no on-chart annotations; shallow or obvious insight.
- 7-8 Annotations, reference lines, shaded regions, or similar devices identify a finding; `final_insight` states magnitude or direction.
- 9-10 Multiple coordinated annotations and an insight-bearing title; publication-quality narrative consistent with Test.

Do not deduct here for overlaps, missing legends, clipped axes, or other rendering failures.

Figure 11: Annotator handbook (3/8): scoring anchors for text-only INSIGHT COMMUNICATION.

5. Visual Quality — rendered PNG

Field: human_visual_quality. Open present figure_path. Judge (i) insight embodiment, (ii) faithfulness to data, and (iii) display legibility. Failure of either gate caps the score at 3.

1 Empty path or missing chart.

1-3 Display failure (clipping, missing multi-series legend, unsuitable chart, generic title); faithfulness failure (misleading truncation, narrative conflict, distribution-hiding aggregation, false causality, temporal cherry-picking, conflicting colors); or generic plot with no finding.

4-6 Passes both gates but does not direct the reader to the finding.

7-8 Passes both gates and uses an annotation, reference line, or statistical overlay.

9-10 Publication-ready, readable, truthful, and supports a specific quantitative conclusion.

Visual calibration examples

≈5/10 Clean default scatter plot with no annotations.

≈8/10 Same plot plus regression line, R^2 , and annotation of the three largest outliers.

≤2/10 Polished time series with a truncated y-axis that makes a 2% change look like 50% (faithfulness failure).

≤2/10 Vertically rotated, mutually overlapping labels that cannot be read (display failure).

Figure 12: Annotator handbook (4/8): scoring anchors and calibration examples for rendered VISUAL QUALITY, including the faithfulness and display gates.

Overall Quality — human_overall_quality

Independent holistic question: Would this result be acceptable if an expert data scientist had conducted the analysis? This is not a weighted average of the five dimensions.

- 1-3 Unacceptable; would fail review.
- 4-6 Adequate but ordinary; basic competence.
- 7-8 Strong; suitable for a team blog.
- 9-10 Exceptional; approaching peer-reviewed publication quality.

Three independent final-insight judgments

insight_factually_correct (yes/no/cannot verify): Check every number and direction against stdout and the figure; use a supplied recomputation script when available. A mismatch, reversal, or self-contradiction is no; insufficient evidence is cannot_verify.

insight_meaningful (yes/borderline/no): Is the finding worth reporting in its domain? Common knowledge, tautologies, and direct value restatements are no; limited value is borderline.

effect_size_reported (yes/no): Does the insight report magnitude via a ratio, difference, percentage, correlation, standardized effect, or similar quantity—not merely significance or qualitative direction?

Complete all three independently. A finding may be correct but meaningless, or meaningful but unverifiable. Even when factual correctness is no, complete the other judgments. Confident random-permutation claims are usually cannot_verify or no.

Figure 13: Annotator handbook (5/8): holistic quality and the three independent final-insight judgments.

Row-level annotation workflow

1. Read the stage-level insight fields.
2. Open present figure path.
3. Independently score all five dimensions from 1-10.
4. Assign human_overall_quality afterward.
5. Complete factual correctness, meaningfulness, and effect-size reporting.
6. Add at most one sentence to annotator_notes for exceptional issues.

Record initials in annotator_id; do not discuss scores with other annotators. Save as human_annotation_<initials>.csv and return independently. In the expanded study, annotators work on a copy of human_annotation_blind_300.csv.

Time budget

Allow approximately 3-5 minutes for each of the first five chains, then 2-2.5 minutes per chain including categorical judgments. Three hundred chains require roughly 10-13 hours. Divide the work into three or four batches with breaks to reduce scoring drift.

Figure 14: Annotator handbook (6/8): row-level annotation workflow and time budget.

Calibration and blinding rules

- Score dimensions independently; e.g., Hypothesis Rigor 9 and Visual Quality 3 is valid.
- Separate dimensions 4 and 5: annotation effort and narrative in code/text versus the rendered PNG. Good annotation code with failed rendering can score high on dimension 4 and low on dimension 5.
- Ignore variable naming and code concision when scoring Visual Quality.
- Preserve blinding: do not inspect `auto_*` model-score columns until human scoring is complete. A gap of ≥ 4 points triggers a check for missed evidence, not a change made merely to agree with the model.
- Random-permutation or content-broken chains usually merit 1-3.
- When uncertain between adjacent scores such as 6 and 7, choose the lower score; upward inflation is more harmful to calibration than downward strictness.

Figure 15: Annotator handbook (7/8): calibration and blinding rules.

Derived measures

- IPM-human agreement: Kendall's τ between `auto_ipm_score` and `human_overall_quality`.
- Per-dimension agreement: Pearson r between each `auto_<dimension>` and `human_<dimension>`.
- Inter-annotator reliability: Krippendorff's α .
- Absolute factual-correctness and meaningfulness rates: per-method proportions of yes judgments.

Malformed, missing, or ambiguous cases

If a chain is malformed, its image is missing, or the rubric does not cover an ambiguity, leave the affected score blank and describe the issue in `annotator_notes` for consistent adjudication. The explicit no-figure rule still applies to Visual Quality: an empty or missing `present_figure_path` receives 1.

Figure 16: Annotator handbook (8/8): derived agreement measures and handling of malformed, missing, or ambiguous cases.

A.9 Quality-Regime System Prompts

The five regimes in § 5.1 (HIGH, MEDIUM-A, MEDIUM-B, LOW, RANDOM-PERMUTATION) are realized by four distinct system-prompt sets — MEDIUM-A and MEDIUM-B share the MEDIUM prompt set but are run under two different random seeds, as explained in § 5.1. The verbatim system prompts below are taken from `data_pipeline/chain_generator.py`; the user prompt template (dataset summary and previous-stage context) is shared across regimes.

HIGH (senior data scientist).

- *Explore*: “You are a senior data scientist with 15 years of experience. Perform a thorough exploratory data analysis. Create 3 diverse, complementary visualizations (distribution, relationship, temporal/categorical); use subplots; examine correlations, outliers, distributions, missing patterns; print descriptive statistics; identify 3 non-obvious patterns worth deeper investigation.”
- *Focus*: “Drill into the MOST non-obvious and impactful pattern. Use advanced techniques: `groupby+agg`, pivot tables, conditional filtering. Create 2 visualizations that dissect the pattern from different angles. Formulate a specific, quantitative, testable hypothesis.”
- *Test*: “Perform rigorous hypothesis testing. Choose the appropriate statistical test (t-test, chi-square, correlation, regression); report p -value, effect size (Cohen’s d , Cramér’s V , etc.), and confidence interval; visualize the test result; mention confounders; state a nuanced conclusion (not just ‘significant’).”
- *Present*: “Create one polished publication-ready chart. Title states the finding as a headline (e.g. ‘Higher X linked to $2.3\times$ more Y’). Use `plt.annotate()` for arrows pointing to key data points; add explanatory annotations; professional palette; subtitle with the key statistic.”

MEDIUM-A / MEDIUM-B (data analyst, two seeds).

- *Explore*: “Create 2 standard charts (e.g. histogram and scatter plot); add titles and axis labels; print basic observations; note 1–2 patterns.”

- *Focus*: “Pick a reasonable pattern from the exploration; use `groupby` or filtering to investigate; create 1 chart; state a hypothesis based on what you see.”
- *Test*: “Use a basic statistical test (e.g. t -test or correlation); report the p -value; create a simple visualization; state whether the hypothesis is supported.”
- *Present*: “Create 1 chart summarizing your main finding; add a descriptive title; use reasonable colors.”

LOW (beginner, one-line tasks).

- *Explore*: “Just pick ONE column and make a histogram. Use `plt.hist()`. Print the mean. That’s it.”
- *Focus*: “Pick any two numeric columns and make a scatter plot. Use `plt.scatter()`. Print correlation. That’s it.”
- *Test*: “Just compare the mean of one column across groups of another. Use `df.groupby().mean()`. Make a bar chart. That’s it.”
- *Present*: “Just make a bar chart or pie chart of a categorical column. Add a title. That’s it.”

RANDOM-PERMUTATION (content-broken control).

- *Explore*: “Write a MINIMAL Python script. ONLY do `print(df.describe())`; then make ONE chart: `plt.plot(df.iloc[:,0])`. Do NOT add titles or labels. Do NOT analyze anything.”
- *Focus*: “Just pick a random column and print value counts. Make a bar chart with `plt.bar()`. No analysis, no labels.”
- *Test*: “Just print the correlation matrix: `print(df.corr())`. Make a heatmap with `sns.heatmap(df.corr())`. Nothing else.”
- *Present*: “Just do `plt.boxplot()`. No title, no labels, no insight.”

A.10 Dataset Tables

This subsection collects the descriptive summaries of the train/val/test referenced from §4: the property-by-property comparison against prior benchmarks (Table 10), source composition (Table 11), per-domain count (Figure 17), and the leave-five-domains-out split sizes (Table 12).

Benchmark	# Datasets	Domains	Multi-step	Cross-domain	Insight Metric	Vision Judge
nvBench (Luo et al., 2021)	750	Narrow	✗	✗	✗ ^a	✗
Prompt4Vis (Li et al., 2025a)	subset	Narrow	✗	✗	✗	✗
Chart-to-Text (Kantharaj et al., 2022)	44k	—	✗	✗	✗ ^b	✗
LIDA exemplars (Dibia, 2023)	ad-hoc	Few	✗	✗	✗	Partial
InsightBench (Sahu et al., 2025)	100	Mixed	Text-only	✗	Text	✗
InsightEval (Zhu et al., 2026)	—	Mixed	Text-only	✗	Text	✗
InsightChain (ours)	285	10*	4-step	✓	5-dim	✓

Table 10: InsightChain vs. prior LLM-visualization and insight-quality benchmarks. ^aRelies on chart-spec matching. ^bRelies on text-based BLEU scores. *Includes 5 held-out domains. To our knowledge, InsightChain is the first benchmark to combine a held-out cross-domain partition with a multi-step chain evaluation and a vision-aware judge.

Source	#	%
TidyTuesday (community-curated)	55	19.3
OpenML (keyword search per domain)	53	18.6
Other public CSV (Kaggle / project pages)	130	45.6
World Bank, FRED & other macro feeds	11	3.9
UCI ML Repository	8	2.8
Yfinance / equity feeds	4	1.4
sklearn classics	1	0.4
Synthetic templates (sparse-domain fallback)	23	8.1
Total	285	100.0

Table 11: Source composition of the InsightChain experiment datasets. Synthetic templates ($n = 23$) are confined to demographics and energy where real public coverage was thinnest after quality filtering.

Split	n	Source
Train	95	5 train dom. $\times$ 19
Val	20	5 train dom. $\times$ 4
In-domain test	35	5 train dom. $\times$ 7
Cross-domain test	135	5 hold-out dom. (all)

Table 12: Split sizes. n refers to the number of sampled datasets. The Train split is used for compilation only, and Val for APO candidate selection. The cross-domain test set is the central testbed for the transfer claim of § 5.3.

A.11 Per-Domain Breakdown

Figure 18 disaggregates the Gemini 3 Flash row of Table 1 by domain. The uncompiled InsightChain outperforms both step-matched baselines across all in-domain training and held-out cross-domain settings. The smallest in-domain gain over single-shot is observed on CLIMATE (+0.44), where long time-series datasets tend to induce more conservative default plotting behavior, while the largest is on SPORTS (+1.85). On the held-out cross-domain evaluation, improvements are more uniform, ranging from +0.77 to +1.16.

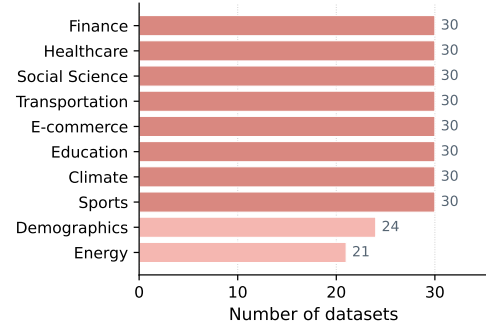


Figure 17: Distribution of datasets across the 10 target domains.

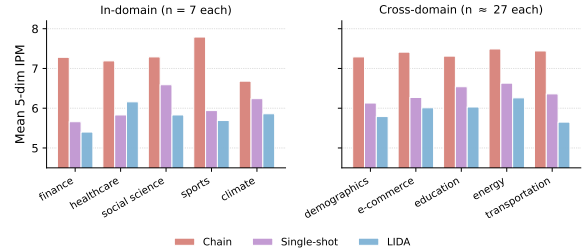


Figure 18: Per-domain mean 5-dim IPM under the Gemini 3 Flash task model. InsightChain wins every in-domain training domain and every cross-domain hold-out.

B Extended Related Work

This appendix complements §2 by connecting InsightChain, IPM, and VG-COPRO to four adjacent lines of work.

B.1 Deep Research and Data-Analysis Agents

Deep-research agents interleave reasoning, tool use, and verification rather than answering in one shot (Yao et al., 2023; Li et al., 2025b). Their treatment of intermediate evidence is especially relevant to InsightChain: Doc-Researcher structures multimodal evidence before reasoning (Dong et al.,

2026), while IE-as-Cache reuses extracted facts across later reasoning steps (Lv et al., 2026a). InsightChain applies this principle specifically to executable analysis: its context $C_{<t}$ (Eq. 2) passes computed findings, not merely dialogue history, between stages. Retaining accumulated context can increase inference cost; AsymSpec addresses this complementary systems problem by allowing a lightweight drafter to access the full context while a larger verifier operates on a compressed view (Liang et al., 2026). Prior work on persistent agent memory (Wu et al., 2025) and step-level rating and refinement (Zhang et al., 2026) provides complementary foundations for state propagation and stage-specific feedback. Data Interpreter similarly plans and repairs code-executing subtasks (Hong et al., 2025), whereas DiscoveryBench evaluates hypothesis search (Majumder et al., 2025) and Multimodal DeepResearcher produces text–chart reports (Yang et al., 2026). InsightChain instead treats a statistically supported visualization as the final deliverable and evaluates both the analytical path and the rendered figure.

B.2 Insight Mining and Insight-to-Visualization

Classical insight-mining systems enumerate and rank predefined patterns (Ding et al., 2019; Ma et al., 2021), while storytelling systems compose those patterns into fact sheets or narratives (Wang et al., 2020; Shi et al., 2021). LLM-based systems make exploration more flexible: InsightPilot places an LLM over an insight engine (Ma et al., 2023), and InsightBench evaluates open-ended insight generation (Sahu et al., 2025). InsightChain combines these directions but grounds each free-form stage in execution: Explore proposes broad candidates, Focus selects hypotheses, Test validates them, and Present renders the result. IPM consequently evaluates whether the final figure communicates the validated finding, rather than merely whether an insight was stated or a chart was produced.

B.3 LLM-as-Judge Evaluation of Visualizations and Multi-Step Reasoning

IPM builds on reference-free, rubric-based LLM evaluation (Liu et al., 2023). Systematic analysis of LLM dialogue evaluators shows why agreement with human judgments must be measured rather than assumed (Zhang et al., 2024a); adversarial analysis of NLG evaluators further motivates our discriminability controls (Chen et al.,

2024). VisEval combines automated checks with a vision-language judge for visualization outputs (Chen et al., 2025), but assesses query-conditioned chart validity and readability. IPM instead asks whether the final chart faithfully communicates a finding established by the preceding executable stages, while four rubric dimensions separately assess those stages. We therefore report human agreement, repeatability, cross-family robustness, and a content-broken RANDOM-PERMUTATION control in §5.1.

B.4 Automatic Prompt Optimization

Automatic prompt optimization searches instructions using scored candidates or textual critiques (Zhou et al., 2023; Pryzant et al., 2023). Adjacent adaptation methods operate below the prompt layer: alignment approaches use model-derived rewards or teacher–student fine-tuning (Zhang et al., 2025, 2024b), while decoding-time steering combines local and global guidance without changing the task prompt (Lv et al., 2025, 2026b). These methods require training or decoding access, whereas prompt optimization applies to black-box APIs. DSPy compilers extend prompt optimization to multi-module programs (Khattab et al., 2024; Opsahl-Ong et al., 2024), but the methods considered in §3.3 rely on text-only feedback. VG-COPRO retains COPRO’s per-predictor search and OPRO-style proposal, while using IPM to supply dimension-specific critique of the full executed pipeline, including the rendered figure. Its leave-one-domain-out worst-case objective (Sagawa* et al., 2020) further selects prompts for transfer rather than average validation performance. Thus, the novelty is not prompt search itself, but adapting its feedback and selection objective to a multi-stage, visually evaluated analysis program.